Could You Be More Sarcastic? A Cognitive Approach to Bidirectional Sarcasm Understanding in Language Models

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Background & Motivation

Language models find it hard to understand sarcasm because:

- It relies on implied meaning, context and speaker intent
- Literal words ≠ intended meaning

Models often detect sarcasm but fail to:

- Generate it naturally
- Remove it while preserving intent

Prior work mostly focused on detection and unidirectional generation **(non-sarcastic → sarcastic)** with no focus on small models.

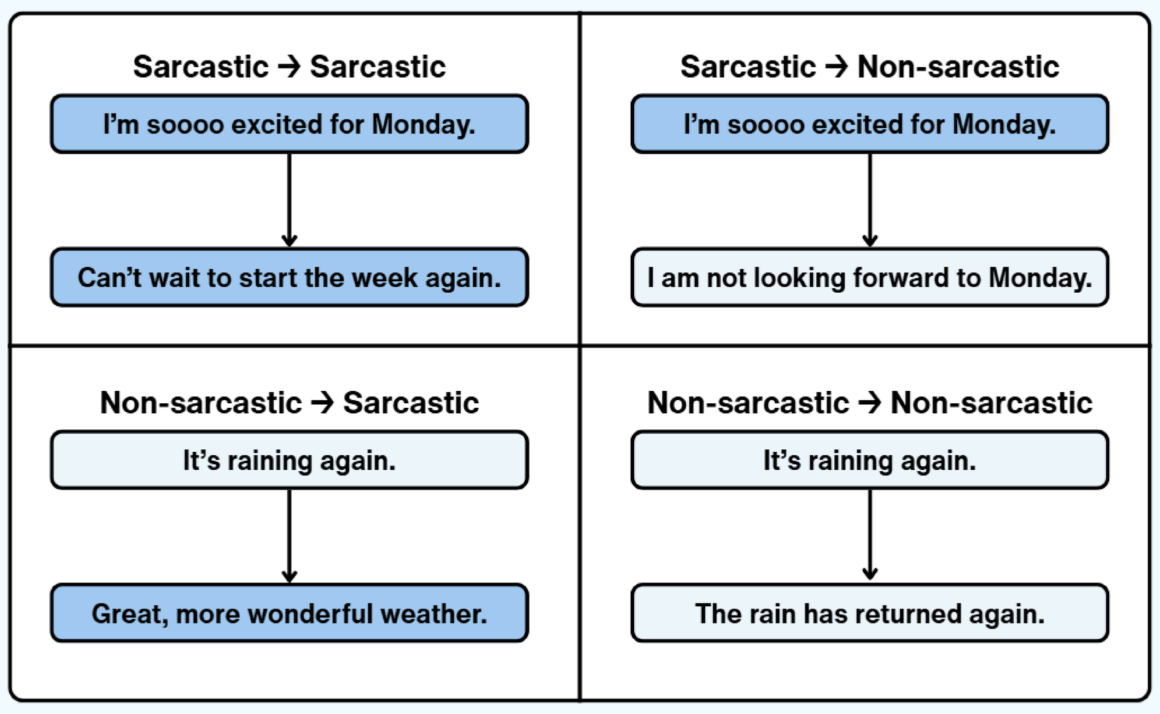
Research Question: Can small language models understand sarcasm as a style and not as a label?

Core Idea & Task

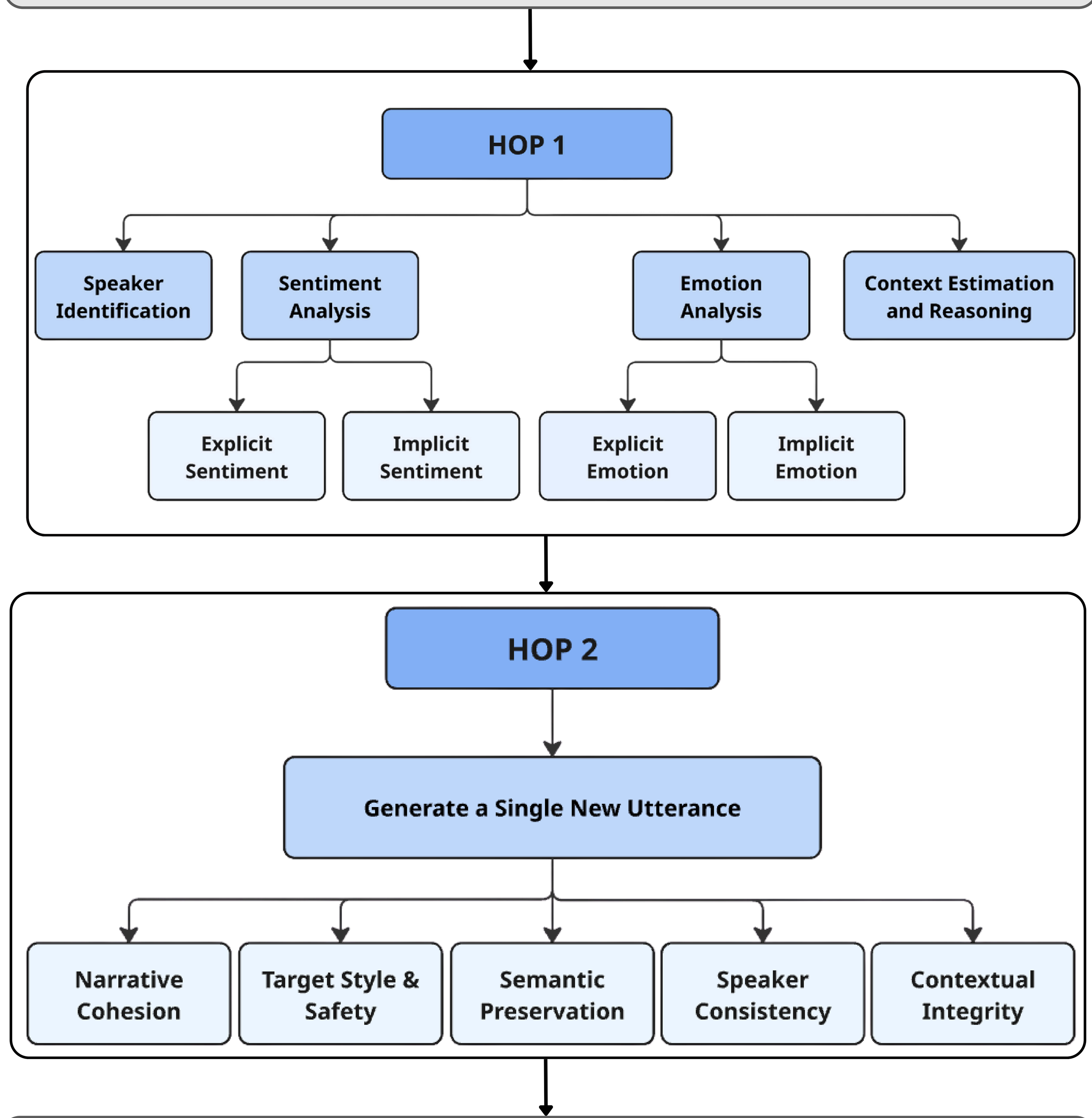
We model sarcasm as a cognitive process using human cognition principles:

- Theory of Mind
- Contextual incongruity
- Implicit vs explicit emotion mismatch

For this, we test both understanding and control in a four-way style transfer.



Dialogue Context:
JOEY: Remember what happened the last time I did an interview for them? I said I write a lot of my own lines, then the writers got mad and made my character fall down the elevator shaft. So who knows what I might say this time.
Reference Utterance:
CHANDLER: If only there was something in your head to control the things you say.



Sarcastic Generated Utterance:
CHANDLER: If only there was a little lock on your tongue to keep the truth from slipping out.
Non-Sarcastic Generated Utterance:
CHANDLER: Maybe if you had more control over what you say, things would go better.

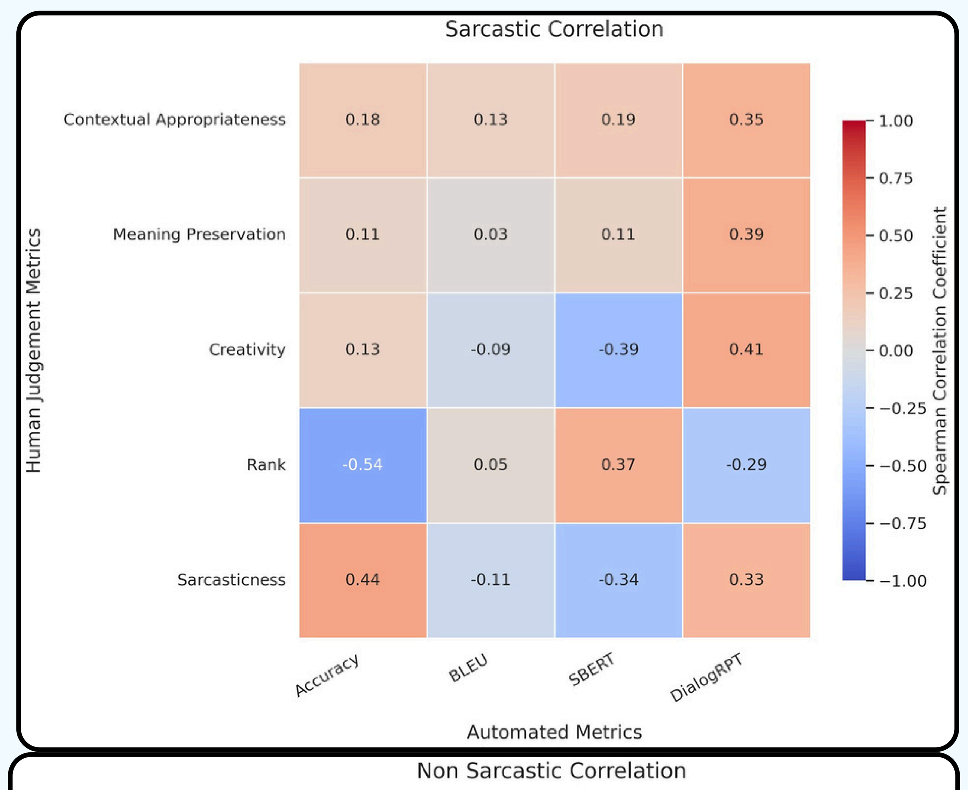
Results & Findings

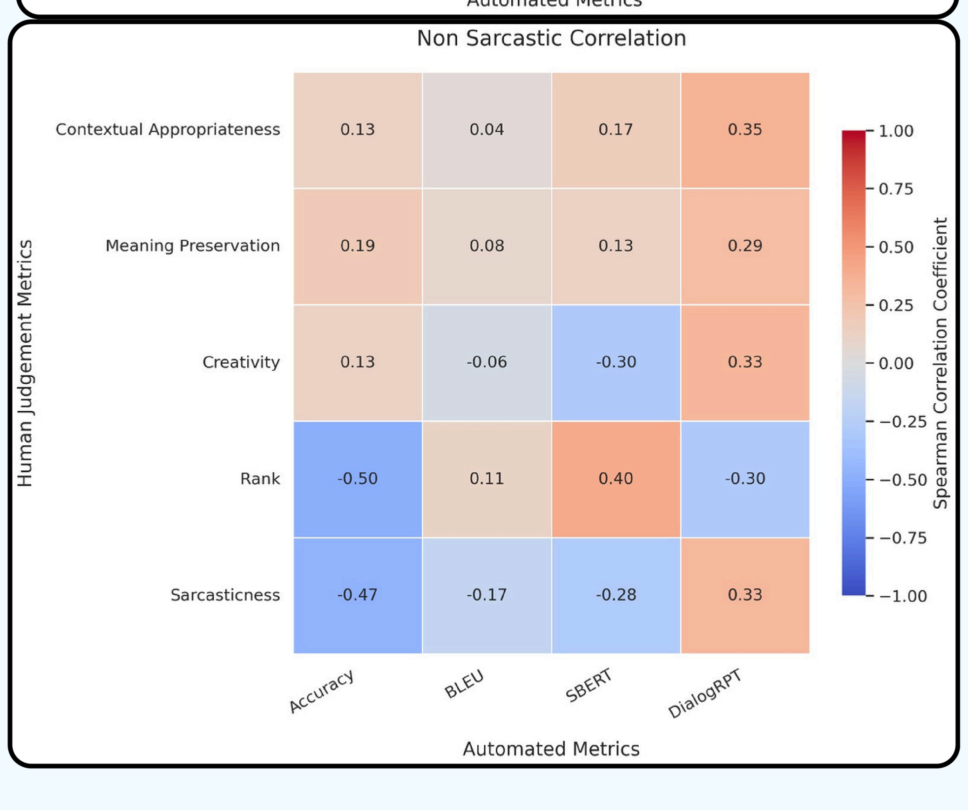
Data: We used the **MUStARD** Dataset with 690 dialogues in a **text-only** setting.
Models:

- **Small Language Models:** Gemma-3 (1B, 4B), LLaMA-3.2 (1B, 3B), Qwen-3 (1.7B, 4B)
- **GPT-4o:** LLM-as-a-judge, upper-bound baseline

Comparative baselines: Zero-shot prompting, few-shot prompting, our framework, few-shot + our framework

- **Need for semantically aware metrics**: Generation metrics like BLEU are limited to lexical overlaps and cannot capture the nuance of sarcasm. Although metrics like DialogRPT are better suited for dialogues, they don't account for subtle changes in intended meaning.
- **Too much context might be overkill**: We utilized multi-hop reasoning to build context that is aligned with human cognitive principles in our strategy. Incorporating few-shot examples into our strategy showed some improvement over baselines in automated metrics but perform sub-optimally in case of LLM and human evaluated metrics
- **Alignment with human judgement**: For a sarcastic context, small language models fail to completely sanitize sarcasm as per our human survey. Some of the limiting factors may also be the distinct perception of sarcasm for every individuals and limitations of tone and intent in text.





Category	Method	Context	Creativity	Meaning	Rank	Sarcasticness.
S→ S	ZS	3.45	3.23	3.32	2.79	3.57
	FS	3.59	3.19	3.13	2.65	3.44
	Ours	3.68	**3.68**	**3.38**	**2.50**	3.70
	Ours+FS	**3.80**	3.64	3.29	2.63	**3.76**
S→ NS	ZS	4.11	2.78	3.57	2.75	**3.15**
	FS	3.83	2.41	3.01	2.60	2.88
	Ours	**4.24**	**2.98**	**3.91**	**2.17**	2.48
	Ours+FS	4.07	2.85	3.68	2.48	2.84
NS→ S	ZS	3.73	3.39	3.01	2.73	2.96
	FS	3.71	3.61	2.80	2.60	3.34
	Ours	**4.01**	**3.81**	**3.51**	**2.21**	3.56
	Ours+FS	3.96	3.69	3.28	2.41	**3.60**
NS→ NS	ZS	3.71	2.55	3.25	2.48	**1.84**
	FS	3.79	2.61	3.17	2.45	1.76
	Ours	**4.17**	**2.77**	**3.97**	**2.11**	1.65
	Ours+FS	3.87	2.61	3.47	2.16	1.79

- Direction-wise Human Evaluation Metrics.
- Sarcastic **(S)**, Non-sarcastic **(NS)**.
- Best performing scores are highlighted in **bold**.

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